

LONDON
SCHOOL *of*
HYGIENE
& TROPICAL
MEDICINE



MSc Project Report 2025-26

Examining the Timeliness of Routine Vaccination During Heatwaves.

Candidate Number: 493736

Page count: 0

Project length: Standard

**Submitted in part fulfilment of the requirements for the degree of MSc in Medical
Statistics**

Date of submission: August 2026

Table of contents

1	Introduction	1
1.1	Background	1
1.2	Aims and Objectives	4
2	Literature review	5
2.1	Vaccine timeliness	5
2.2	Heatwaves	7
2.3	Heatwaves and public health	8
3	Materials and Methods	10
3.1	Data	10
3.2	Study population	11
3.3	Outcome variables	11
3.4	Exposure variables	12
3.5	Covariates	14
3.6	Statistical analysis	15
4	Results	18
4.1	Study population	18
4.2	Heatwave exposure	19
4.3	Heat Index exposure	19
4.4	Vaccine timeliness	19
4.5	Heat exposure and vaccine timeliness	19
4.6	Timeliness statistics by threshold	19
5	Discussion	20
	References	21
	Appendix	32

Abstract

All project reports should include a structured Abstract, not exceeding 300 words, on a single standalone page just after the Contents. This should appear before the main body of the project report (which will start with the introduction).

The Abstract may be structured into four key sections: - Background – summarising the problem being considered. - Methods – describing how the study was performed. - Results – listing the salient results. - Conclusions – stating the principal conclusions.

Acknowledgements

Your project report should include an Acknowledgements section, before the Introduction to your work proper. Detailed guidance on what to put in the Acknowledgements section is given in Section 12.1

1 Introduction

1.1 Background

Globally, extreme weather events such as floods, cold waves, droughts and heatwaves have become increasingly frequent and intense(1). These events disrupt multiple sectors including energy infrastructure, agriculture, food production, healthcare and livelihoods, resulting in substantial social and economic losses (2,3). Consistent with these changes, global mean near-surface temperatures have risen by approximately 1.4–1.5°C above pre-industrial (1850–1900) levels, with the five-year average temperature during 2020–2025 being the highest on record and 2024 recorded as the warmest year globally (4). This continued warming is largely driven by increasing greenhouse gas emissions, together with changes and interactions in atmospheric and oceanic circulation (4,5). One of the most direct consequences of rising global temperatures is the increasing occurrence of heatwaves, commonly defined as periods of unusually high temperatures that persist for several consecutive days. Climate projections indicate that both the frequency and intensity of heatwaves will continue to increase unless substantial reductions in greenhouse gas emissions are accompanied by investments in climate adaptation and resilient infrastructure (5).

Heatwaves have important implications for both population health and the delivery of healthcare services. Extreme heat places additional pressure on health systems through increased demand for healthcare, disruptions to essential services, and challenges in maintaining critical infrastructure. At the individual level, heat exposure has been associated with an increased incidence of heat-related illnesses and injuries, particularly among older adults, prolonged recovery among hospitalized patients, increased in-patient mortality, and excess mortality across many regions of the world (6–11). Heatwaves may also compromise the storage and transport of temperature-sensitive medical products, including vaccines, medicines and medical equipment, by placing additional strain on cold-chain systems (12,13). These impacts increase pressure on already constrained healthcare systems and have the potential to disrupt the delivery of essential public health services, including routine childhood immunization.

Nigeria has experienced several notable heatwave events in recent decades, including those occurring in 2002, 2019, 2020, 2023 and 2024 (14–16). These events have been linked to broader climatic changes, compounded by rapid urbanisation, deforestation and land-use change. Their

impacts have been felt across multiple sectors. In agriculture, the 2024 heatwave was associated with an estimated 20% decline in agricultural production in a country where food production depends heavily on rainfall (16,17). Heatwaves have also increased pressure on Nigeria's energy infrastructure through higher cooling demand, contributing to electricity shortages and national grid failures (17). In the health sector, they have been associated with increased occurrence of heat-related illnesses such as heat stroke and meningitis, while environmentally they have contributed to increased wildfire risk and the progression of desertification, particularly in northern Nigeria (17,18).

Nigeria's National Programme on Immunization (NPI) was established in 1978 to provide routine immunization services for children. Over the years, the programme has expanded to include additional vaccine antigens in line with the country's Expanded Programme on Immunization (EPI). According to the current national immunization schedule (Table 1.1), a child is considered fully immunized upon receiving one dose of *Bacillus Calmette–Guérin* (BCG), the hepatitis B birth dose, four doses of oral polio vaccine (OPV), two doses of inactivated polio vaccine (IPV), three doses each of the pentavalent, pneumococcal conjugate (PCV) and rotavirus vaccines, two doses of measles-containing vaccine (MCV), and one dose each of the yellow fever and meningitis vaccines [(19)].

Despite substantial progress over recent decades, routine childhood immunization coverage in Nigeria remains below both national and global targets and continues to be characterised by marked geographical and socioeconomic disparities (20,21). Although the proportion of children receiving all basic vaccinations increased from 13% in 2003 to 39% in 2024, coverage remains considerably lower in northern Nigeria than in the southern regions (19,22,23). These inequalities have been associated with several factors, including place of delivery, maternal education, household wealth, and urban–rural residence (19,23,24). Consequently, Nigeria continues to account for a substantial proportion of the global burden of undervaccinated children (21). Persistent challenges facing the national immunization programme include supply chain constraints, shortages of healthcare personnel and funding, limited geographical access to health facilities, and vaccine hesitancy driven by sociocultural and religious factors (21,25). More recently, the COVID-19 pandemic further disrupted routine immunization services, slowing progress towards improving vaccination coverage and access (26).

Although vaccination coverage remains the most commonly reported indicator of immunization

Table 1.1: Nigeria's National Programme on Immunization schedule.

Minimum target age	Vaccine
At birth	BCG
	OPV 0 (Birth dose)
	Hepatitis B birth dose
6 weeks	Pentavalent (DPT, Hepatitis B, HiB) 1
	Pneumococcal Conjugate Vaccine 1
	OPV 1 (1st dose)
	Rotavirus 1
	IPV1
10 weeks	Pentavalent (DPT, Hepatitis B, HiB) 2
	Pneumococcal Conjugate Vaccine 2
	OPV 2 (2nd dose)
	Rotavirus 2
14 weeks	Pentavalent (DPT, Hepatitis B, HiB) 3
	Pneumococcal Conjugate Vaccine 3
	OPV 3 (3rd dose)
	Rotavirus 3
	IPV2
6 months	Vitamin A (1st dose)
9 months	Measles 1st dose (MCV1)
	Yellow Fever
	Meningitis Vaccine
12 months	Vitamin A (2nd dose)
15 months	Measles 2nd dose (MCV2)

program performance, it does not capture whether vaccines are administered at the recommended ages (27,28). Vaccination timeliness refers to the administration of vaccines within the age windows specified by the national immunization schedule (29). Unlike coverage, which only measures whether a child has eventually received a vaccine, timeliness distinguishes between early, timely and delayed vaccination. This distinction is important because vaccines administered too early may result in reduced vaccine effectiveness owing to interference from maternal antibodies or an immature infant immune system (30,31), while delayed vaccination prolongs children's susceptibility to vaccine-preventable diseases during periods of greatest risk and has been associated with disease outbreaks even in populations with relatively high vaccination coverage (32). Importantly, there is increasing recognition that routine immunization programs should evaluate both vaccination coverage and vaccination timeliness. Recent studies have therefore advocated for reporting age-specific vaccination coverage and timeliness indicators to provide a more complete assessment of program performance and to better identify populations at increased risk of vaccine-preventable disease outbreaks (33,34).

Previous studies have extensively investigated socioeconomic determinants of vaccination timeliness (35), while research on climate-sensitive determinants has remained limited. Existing studies examining heat and health have primarily focused on mortality, morbidity and healthcare utilisation, with little evidence on routine childhood vaccination and none on vaccination timeliness in Africa. As climate change continues to increase the frequency and intensity of heatwaves, understanding whether these events influence childhood immunization has important implications for vaccination programs and climate adaptation planning (36,37).

1.2 Aims and Objectives

The aim of this study is to examine the association between heatwave exposure and the timeliness of routine childhood vaccination among children aged 0-36 months in Nigeria.

The specific objectives are as follows:

1. Estimate the spatial and temporal distribution of heatwave occurrence across Nigeria during 2021–2024.
2. Estimate the timeliness of the BCG, Pentavalent 1–3 and MCV1 vaccination.
3. Assess the association of heatwaves and vaccine timeliness while adjusting for selected socioeconomic and demographic characteristics.

2 Literature review

2.1 Vaccine timeliness

Several country-specific studies, multi-country analyses, and systematic reviews have examined the timeliness of routine childhood vaccination using nationally representative survey data, primarily from the Demographic and Health Surveys (DHS) and Multiple Indicator Cluster Surveys (MICS) (28,33–35,38–42). The largest and most recent systematic review synthesised evidence from 224 studies conducted over four decades (1978-2024) across low- and middle-income countries (33). Despite this growing body of evidence, systematic reviews have highlighted important methodological limitations and substantial heterogeneity in the approaches used to assess vaccination timeliness (33,35).

Across all studies, vaccination timeliness has generally been measured using one of two approaches. The first categorizes vaccination into early, timely and delayed administration or delayed versus timely administration(33,41,42), while the second treats delay as a continuous time-to-event outcome, typically defined as the number of days between the recommended age of vaccination and the observed vaccination date (34,38,40). Although both approaches are widely used, no universally accepted definition of vaccine delay exists(33). Consequently, vaccination delay has been defined using a variety of thresholds based on the time elapsed since the recommended age for vaccination. Common thresholds include four days, one week, two weeks, four weeks (28 days), one month, two months, and ninety days after the scheduled vaccination age (33,35). Whether vaccine timeliness is analysed as a discrete or continuous outcome was found to have important implications for the choice of statistical models. Studies using categorized measures of timeliness employed logistic regression models, whereas those analyzing vaccination delay as a continuous time-to-event outcome typically used survival analysis methods, including the Cox proportional hazards model (35).

Another important source of methodological variation relates to the age range of children included in vaccine timeliness studies. A recent systematic review identified 35 different age ranges across 57 studies, although children aged 12–23 months were the most commonly studied population (35). Overall, the review found substantial inconsistency in age eligibility, with studies including children ranging from those younger than one year to those aged five years and above, while some studies did not clearly define the age range of participants. Such variation may

influence reported estimates of vaccine timeliness, particularly for vaccines administered later during infancy such as the second measles containing vaccine, as younger children may not yet have reached the recommended age for all scheduled vaccinations, whereas older children have had a longer opportunity to receive delayed vaccines.

Studies have consistently identified several determinants of vaccination timeliness. The determinants considered across most studies are generally divided into child-level factors such as child's sex, birth order, place of birth, birth year, maternal and paternal factors such as age, level of education and occupation, household level factors such as religion, ethnicity, wealth index, residence type, travel times to facilities and health system factors such as health facility availability, accessibility and staffing(33,35). Across major studies, children living in rural areas, those from poorer households, and those whose mothers had little or no formal education were consistently more likely to experience delayed vaccination (28,33,42). These findings suggest that socioeconomic inequalities remain an important determinant of vaccine timeliness and highlight the need for interventions targeted towards disadvantaged populations (28). Further, vaccination was generally more timely for birth doses and earlier childhood vaccines, while delays increased for vaccines scheduled later in infancy (34,42). A recent analysis of more than 90 DHS and MICS surveys reported median timely vaccination coverage of approximately 79% for birth doses, declining to 70% at six weeks, 59% at ten weeks, 50% at fourteen weeks, 56% at nine months and 50% at fifteen months (34). This pattern suggests that children become progressively more likely to experience delayed vaccination as they progress through the routine immunization schedule. Beyond individual and household characteristics, several studies have attributed delayed vaccination to health system disruptions such as Covid-19 (43), health-system constraints, including vaccine stock-outs, inadequate supply chains, shortages of healthcare personnel, limited access to health facilities, and civil unrest or insecurity in some settings[xxx]. These factors may disrupt routine immunization services and reduce access to timely vaccination, particularly in resource-constrained settings.

Studies of vaccine timeliness have also differed in how they handled children with missing vaccination dates, which are required to calculate vaccination delay and timeliness. Some studies excluded these children from the analysis, whereas others used survival analysis techniques that accounted for incomplete information (33). For example, left censoring was used for children whose mothers reported that the vaccine had been received before the interview but without a recorded date, while right censoring was applied to children who had not yet received the vaccine by the time of the interview. However, few studies incorporated both left and right censoring.

These inconsistencies in age groups used, definitions and operationalization of vaccine delay, adjusted predictor variables and modelling approaches complicate comparisons and harmonization of vaccine timeliness estimates across studies and countries.

2.2 Heatwaves

Heatwaves are among the most widely studied climate extremes because of their increasing frequency, intensity and duration under climate change. Although there is no universally accepted definition of a heatwave, most definitions describe a heatwave as a period of abnormally high temperatures persisting for several consecutive days relative to the local climatology(44,45). Earlier studies often relied on absolute temperature thresholds (for example, daily maximum temperatures exceeding a fixed value), while more recent approaches favour percentile- or quantile-based definitions that account for local climatic conditions and population adaptation, such as temperatures exceeding the 90th, 95th or 99th percentile of the historical temperature distribution for a given location and time of year (44,46,47). Other studies have incorporated both temperature intensity and duration by defining heatwaves using moving averages of daily maximum temperatures or indices that combine daytime and nighttime temperatures. The evolution from fixed thresholds to location-specific and duration-sensitive definitions has substantially improved comparability across climatic regions and has enabled more robust assessments of heatwave occurrence under changing climatic conditions (44,47,48).

Alongside advances in heatwave definitions, considerable progress has been made in attributing individual heatwave events to human-induced climate change (44,46). Early attribution studies primarily compared observed temperatures with long-term climatological averages, whereas modern attribution frameworks combine observational records, climate model simulations and extreme value statistics to quantify changes in both the probability and intensity of extreme events. Event attribution is now commonly undertaken using probabilistic approaches such as the World Weather Attribution (WWA) framework, which compares the probability of an event occurring in the present-day climate with that in a counterfactual climate without human influence(46). These approaches frequently employ generalized extreme value (GEV) models fitted to annual temperature maxima to estimate return levels and return periods, allowing rare events to be characterised using exceedance probabilities or high quantiles of the fitted extreme value distribution. The resulting metrics, including probability ratios and fractions of attributable risk, provide quantitative evidence of the contribution of human-induced climate change to observed heatwave events. Recent methodological developments have highlighted the importance of

accounting for non-stationarity, model uncertainty and physical consistency when estimating return levels and attributing extreme heat events under a warming climate(44,46,49). Recent work has further argued that traditional attribution methods require refinement to better account for compound climate processes, observational uncertainty and evolving background climatic conditions, particularly when projecting future heatwave risks (44).

Heatwave occurrence has increased across Africa over recent decades, although important regional heterogeneity exists in both exposure and vulnerability. Compared with many high-income settings, populations in sub-Saharan Africa face disproportionately higher risks owing to rapid urbanisation, widespread poverty, limited adaptive capacity and weaker health infrastructure (50). Evidence suggests that heatwaves are becoming more frequent, longer lasting and more intense across much of the continent, with urban areas experiencing particularly severe warming because of the urban heat island effect (18,51). Within low- and middle-income countries, heatwaves have been linked to increased demand for healthcare, reduced labour productivity, food insecurity and disruptions to essential services, highlighting the need for locally appropriate heat adaptation strategies (2,16,37,52). In Nigeria, recent analyses have documented increasing heatwave frequency and intensity, particularly across the northern regions, with projected future increases under continued climate warming (15,16). These events have been associated with impacts on agriculture, electricity demand, environmental degradation and public health, while rapid urbanisation and land-use change are expected to further amplify future heat exposure (16–18).

2.3 Heatwaves and public health

The health impacts of heatwaves are now well established. Numerous epidemiological studies have demonstrated associations between extreme heat and increased all-cause mortality, cardiovascular and respiratory disease, renal disease, adverse pregnancy outcomes, heat stroke and hospital admissions, with older adults, young children and individuals with chronic illnesses being particularly vulnerable (6–11). Beyond direct physiological effects, heatwaves may also disrupt healthcare delivery through increased pressure on health facilities, interruptions to electricity supply and cold-chain infrastructure, and reduced healthcare utilization (12,13). These disruptions are particularly relevant for routine childhood immunization programs, where reliable vaccine storage and timely attendance at health facilities are essential (53,54). Existing studies examining climate and vaccination have largely focused on vaccine storage, cold-chain integrity, seasonality and healthcare access, while relatively few have investigated whether

exposure to extreme heat influences vaccination coverage or the timeliness of routine childhood immunization (36,37,55).

To date, evidence directly linking heatwave exposure to vaccine timeliness is lacking, particularly in low- and middle-income countries and sub-Saharan Africa. This represents an important knowledge gap given the increasing frequency of heatwaves and the importance of timely vaccination in ensuring optimal protection against vaccine-preventable diseases.

3 Materials and Methods

3.1 Data

3.1.1 DHS Data

The 2024 Nigeria DHS survey (NDHS) was conducted between December 2023 and May 2024 by the National Population Commission (NPC) and was designed to provide nationally representative estimates for Nigeria as a whole, as well as for urban and rural areas, the six geopolitical zones, and each of the 36 states and the Federal Capital Territory (FCT). The 2024 NDHS employed a stratified two-stage cluster sampling design based on the updated cartographic sampling frame prepared for Nigeria's Population and Housing Census. The sampling frame excluded institutional populations such as persons in hotels, barracks, and prisons. In the first stage, enumeration areas (EAs), which served as primary sampling units or clusters, were selected with probability proportional to size within each of the 74 sampling strata defined by state and place of residence (urban/rural). A total of 1,400 clusters were selected, comprising 701 urban and 699 rural clusters. In the second stage, a household listing was conducted in each selected cluster, after which 30 households were selected using systematic sampling, yielding the intended sample of approximately 42,000 households(19).

Data for this study were obtained from the Women's Questionnaire of the 2024 NDHS. Women aged 15–49 years who were usual members of selected households or who had spent the night preceding the survey in the selected household were eligible for interview. The questionnaire collected information on respondents' sociodemographic characteristics, reproductive history, maternal and child health, family planning, and other health-related indicators. In total, 39,553 women were eligible for interview (19,166 from urban areas and 20,387 from rural areas), and a response rate of 99% was achieved nationally as well as in both urban and rural areas.

Information on childhood vaccination was obtained for children born in the three years preceding the survey, and alive at the time of the survey. For each eligible child, vaccination information, including receipt of each vaccine antigen and the corresponding date of administration, was extracted from the child's vaccination card where available. In the absence of a vaccination card, vaccination status was determined based on maternal recall. According to the 2024 NDHS, vaccination cards were seen for 50.2% of children aged 12–23 months and 34.4% of

children aged 24–35 months, indicating that maternal recall constituted an important source of vaccination information for a substantial proportion of children(19).

3.1.2 Temperature data

Data on daily mean 2-metre air temperature for the period 1940–2024 and daily mean 2 metre dewpoint temperature for the period 2020–2024 were obtained from the European Centre for Medium-Range Weather Forecasts (ECMWF) ERA5 reanalysis via the Climate Data Store(56,57). The data are provided on a regular $0.25^{\circ} \times 0.25^{\circ}$ latitude–longitude grid (approximately 27.8 kilometres squared). The data were extracted for a spatial bounding box enclosing Nigeria (west: 2.676932°E , south: $4.0690959^{\circ}\text{N}$, east: $14.678014^{\circ}\text{E}$, north: $13.885645^{\circ}\text{N}$) and retrieval was carried out programmatically in R using the `ecmwfR` package(58), which provides an interface to the ECMWF data services and the Climate Data Store API. The downloaded files were stored on secure local storage in NetCDF format, with one file saved for each year to facilitate efficient management and analysis.

3.2 Study population

The analysis was restricted to children born in the three years preceding the 2024 Nigeria Demographic and Health Survey. Children who had died before the survey were excluded since the DHS birth recode only collects vaccination information for living children and children who had missing information on date of birth were also excluded. For each vaccine antigen, the analysis included only children who had reached the recommended age of eligibility for that vaccine at the time of interview. For example, analyses of the first dose of the measles-containing vaccine (MCV1), which is scheduled at 9 months of age, were restricted to children aged at least 9 months. Therefore, the analytical sample differed slightly across vaccine antigens.

3.3 Outcome variables

The primary outcomes were delays in receipt of five key routinely recommended childhood vaccine antigens: Bacillus Calmette–Guérin (BCG), the first, second and third doses of the pentavalent vaccine (Penta1, Penta2 and Penta3), and the first dose of the measles-containing vaccine (MCV1). The pentavalent vaccine protects against the combination of diphtheria, tetanus,

pertussis, hepatitis B and Haemophilus influenzae type b infections. According to Nigeria's Expanded Programme on Immunization (EPI) schedule(22), BCG is recommended at birth, Penta1 at 6 weeks, Penta2 at 10 weeks, Penta3 at 14 weeks, and MCV1 at 9 months. These antigens were selected because they represent key milestones in the infant immunization schedule and are widely used as indicators of the performance of routine immunization programmes during the first year of a child's life.

For each antigen, the recommended vaccination date was calculated by adding the scheduled age at vaccination to the child's date of birth. Timeliness was assessed at 7 days, 14 days, 21 days and 28 days. Although no universally accepted definition of vaccination delay exists (35), this study defines vaccine delay as receipt of a vaccine more than 28 days after its recommended due date, so that vaccines administered within 28 days of the scheduled date were classified as timely (59). This definition was adopted across all the antigens considered for analysis

3.4 Exposure variables

The data on daily mean 2-metre air temperature and mean 2-metre dewpoint temperature from ERA5 were used to derive heatwave and heat index exposure variables. The ERA5 temperature data were spatially linked to the 2024 Nigeria DHS clusters using the geographic coordinates provided in the survey. Following DHS recommendations for incorporating spatial covariates while accounting for the random displacement of cluster coordinates [xxx] (The DHS applies random displacement to cluster coordinates to protect confidentiality, with displacement distances of up to 2 km for urban clusters, 5 km for 99% of rural clusters, and 10 km for 1% of rural clusters). In this study, cluster locations were matched to ERA5 grid cells using a 10-kilometre buffer radius. Given that the ERA5 spatial resolution is approximately 0.25° (approximately 27 km), which is substantially larger than the maximum DHS displacement distance, coordinate displacement would be expected to have limited influence on the assigned temperature exposure. Therefore, the spatial linkage would generally correspond to assigning each DHS cluster the temperature series from the ERA5 grid cell containing its displaced centroid. Following this spatial linkage, all subsequent analyses were conducted using temperature time series corresponding to each DHS cluster location.

3.4.1 Heatwave exposure variable

Heatwaves were defined following the probabilistic extreme event attribution framework (10,44,46). Specifically, a heatwave was defined as a 5-day rolling average temperature exceeding the estimated local 75th percentile derived from a fitted GEV model.

To capture sustained periods of unusually high temperatures and account for the cumulative effects of heat exposure, a five-day rolling mean of daily 2-metre air temperature (Tx5x index) was calculated for each DHS cluster location for the period January 1940 to May 2024. The resulting temperature time series was divided into a baseline period (January 1940–December 2019) and an analysis period (January 2020–May 2024). The baseline period was used to estimate cluster-specific extreme temperature thresholds, while the analysis period was used to identify heatwave events occurring during the period covered by the NDHS. For each DHS cluster, the annual maximum of the Tx5x index was extracted from each year of the baseline period, yielding 80 annual maxima. These annual maxima were modelled using a non-stationary Generalized Extreme Value (GEV) distribution in which the location parameter varied linearly with the Global Mean Surface Temperatures (GMST) to account for long-term warming trends(44,46):

$$M_{t,x} \sim GEV(\mu, \sigma, \zeta)$$

$$\mu = \beta_0 + \beta_1 \cdot GMST$$

where $M_{t,x}$ denotes the annual maximum Tx5x value at cluster x in year t , and $GMST$ denotes the Global Mean Surface Temperature in year t . The model was fitted using the `gamlss` and `gamlssx` packages in R(60,61).

The fitted GEV models were subsequently used to estimate Z_4 , the 4-year return level for each DHS cluster, representing the temperature expected to be exceeded, on average, once every four years under the fitted non-stationary climate model. If $F(\cdot)$ denotes the fitted GEV cumulative distribution function(62):

$$F(x) = \exp(-[1 + \zeta(\frac{x - \mu}{\sigma})]_+^{-1/\zeta})$$

Then, the T -year return level satisfies

$$P(M_{t,x} > Z_T) = \frac{1}{T}; F(Z_T) = 1 - \frac{1}{T}$$

For a 4-year return level, $F(Z_4) = 1 - 1/4 = 0.75$ and hence $Z_4 = F^{-1}(0.75)$ which is given by:

$$Z_4 = \mu + \frac{\sigma}{\zeta} [(-\log_e(0.75))^{-\zeta} - 1], \zeta \neq 0$$

Each Tx5x observation during the analysis period (January 2020–May 2024) was then compared with the corresponding DHS cluster-specific return level. Days for which the Tx5x index exceeded the return level (Z_4) were classified as heatwave days, whereas all remaining days were classified as non-heatwave days. This produced a daily binary heatwave indicator for every DHS cluster and each observation was classified as heatwave-exposed if at least one heatwave day occurred within a 15-day exposure window centered on the scheduled vaccination date (7 days before to 7 days after).

3.4.2 Heat Index exposure variable

The heat index was computed using the 2-metre air temperature and the 2-metre dewpoint temperature. The `weathermetrics` package (63) was then used in the computation of the heat index using the heat Index calculation algorithm of the U.S. National Weather Service (NWS) (64)

The heat index was categorized according to NWS guidelines as shown:

Classification	Heat Index (°C)
Tolerable	<26.7°C
Caution	26.7°C–32.2°C
Extreme Caution	32.2°C–39.4°C
Danger	39.4°C–51.1°C
Extreme Danger	≥51.7°C

3.5 Covariates

In this study, the covariates utilized were informed by a recent systematic review of factors associated with vaccine timeliness in low- and middle-income countries over 10 years (2007-2017)(35), and availability of the covariates in the DHS data. These comprised child, maternal, household and geographical characteristics that have previously been associated with vaccination

timeliness. Continuous variables were categorized where appropriate to improve interpretability and to ensure consistency with standard DHS reporting. Maternal age at the child's birth was grouped into seven five-year age categories, while heat index was classified according to the United States National Weather Service heat index categories into Normal, Caution, Extreme Caution, Danger and Extreme Danger. Place of residence, place of delivery, child's sex, maternal education, household wealth index, perceived travel time to the nearest health facility, and geographical zone were analysed using the standard DHS classifications. Birth order was retained as a continuous variable. The covariates included in the analysis and their corresponding types are presented in Table 3.1.

Table 3.1: Description of predictor variables included in the analysis.

Variable	Description	Type
Gender	The child's gender	Binary
Delivery	The place of delivery categorized as either institutional or home delivery	Binary
Birth order	The child's birth order	Integer
Education	The highest level of education attained by the mother. Categories include: No education, primary, secondary and higher education.	Categorical
Age-at-birth	The age of the mother at birth	Categorical
Geozone	Geopolitical zone. Nigeria is divided into six of these zones: North-West, North-East, North-Central (Middle Belt), South-West, South-East, and South-South	Categorical
DHF	The distance to the nearest health facility in minutes	Categorical
Residence	The residence, categorized as rural or urban	Binary
Wealth	The household wealth index categorized into 5 quintiles (poorest, poorer, middle, richer, richest)	Categorical

3.6 Statistical analysis

The primary outcome of interest was the age at vaccination, measured as the number of days from birth until receipt of the vaccine under consideration. For a specific antigen, let T_i denote the age at vaccination for child i . Since vaccination status differed across children at the time of interview, three forms of observations were present. For children whose vaccination date was recorded on the vaccination card, the event time T_i was observed exactly. For children whose mothers reported that the child had received the vaccine but the exact vaccination date was unavailable (typically because no vaccination card was available), the event time was interval censored such that, if L_i denotes the recommended age for the vaccine (Table 1.1) and U_i

denotes the child's age at interview, then the true vaccination time satisfies $L_i < T_i \leq U_i$. Children who had not received the vaccine by the interview date were treated as right censored, such that $T_i > U_i$. The age at vaccination was modelled using a parametric accelerated failure time (AFT) model assuming a Weibull distribution:

$$T_i \sim \text{Weibull}(\lambda_i, \kappa),$$

where κ denotes the Weibull shape parameter and λ_i is the individual-specific scale parameter. The AFT model assumes a log-linear relationship between the expected survival time and the explanatory variables,

$$\log(\lambda_i) = \beta_0 + \beta_1 H_i + \mathbf{X}_i^\top \gamma,$$

where β_0 is the intercept, H_i is the binary heatwave exposure indicator (1 = exposed, 0 = not exposed), $\mathbf{X}_i$ is the vector of adjustment covariates, and γ is the corresponding vector of regression coefficients.

To investigate whether the severity of heat exposure influenced vaccination timing, a second model replaced the binary heatwave indicator with the categorical heat index exposure variable,

$$\log(\lambda_i) = \beta_0 + \sum_{k=2}^K \beta_k I(HI_i = k) + \mathbf{X}_i^\top \gamma,$$

where HI_i denotes the heat index category for child i , $I(HI_i = k)$ is an indicator function for the k^{th} heat index category, and the lowest heat category served as the reference level.

Parameter estimation was performed by maximum likelihood. For an individual child, the likelihood contribution depended on the censoring mechanism. Let $f(\cdot)$, $F(\cdot)$ and $S(\cdot) = 1 - F(\cdot)$ denote the Weibull probability density, cumulative distribution and survival functions, respectively. The individual likelihood contribution was constructed as follows:

$$L_i(\theta) = \begin{cases} f(T_i), & \text{if the vaccination date was observed,} \\ F(U_i) - F(L_i), & \text{if the vaccination time was interval censored,} \\ S(U_i), & \text{if the child was right censored.} \end{cases}$$

The full likelihood was then obtained by taking the product of the individual likelihood contributions,

$$L(\theta) = \prod_{i=1}^n L_i(\theta),$$

where θ denotes the vector of Weibull and regression parameters. Maximum likelihood estimates were obtained using the `svysurv()` function from the `survival` package in R, which fits Weibull accelerated failure time models while accommodating exact, interval-censored and right-censored observations within a unified likelihood framework that handles the multi-stage sampling design and sampling weights. Regression coefficients from the AFT model were exponentiated and interpreted as event time ratios. An event time ratio greater than one indicates a longer expected time to vaccination (greater delay), whereas an event time ratio less than one indicates a shorter expected time to vaccination (earlier vaccination), after adjusting for the selected covariates.

3.6.1 Statistical software

All analyses were conducted using R version 4.5.1(65). Data wrangling, manipulation and visualization were performed using the `tidyverse` suite of packages(66). Data retrieval and management were carried out using `ecmwf`(67), `haven`(68), and `terra`(69), while spatial processing and analysis were performed using the `terra` and `sf` packages(69,70). Extreme value modelling was undertaken using `gamlss` and `gamlssx`(71,72), computation of the heat index was done using the `weathermetrics` package(63) and model fitting was conducted using the `survey` (73,74) package. Reproducible reports and tables were generated using `rmarkdown` and `kableExtra`(75,76).

4 Results

4.1 Study population

The 2024 Nigeria Demographic and Health Survey (NDHS) included a total of 104,557 children. After restricting the sample to children born within the three years preceding the survey, 15,993 children remained. Further restricting the sample to children who were alive at the time of the survey resulted in a final sample of 14,916 children. The number of children eligible for analysis for each antigen, based on the recommended vaccination schedule from the National Programme on Immunization (NPI), is presented in Table 4.1 together with unweighted statistics across selected covariates. The number of children eligible for antigen-specific analysis decreased progressively across the vaccination schedule, from 14,896 children eligible for BCG at birth to 10,837 children eligible for MCV1 at 9 months.

The distribution of the characteristics was similar across the antigen-specific datasets. Approximately 79% of children had a reported travel time of less than 30 minutes to the nearest health facility across all antigens. The distribution of children across wealth index categories was relatively uniform, although the largest proportions were in the poorest (25%) and poorer (20%) quintiles, while smaller proportions were observed in the richer (19%) and richest (17%) quintiles. Maternal age at birth was concentrated among mothers aged 20–34 years. The North West geopolitical zone accounted for the largest proportion of children across the antigen-specific datasets. The number of children residing in rural areas was higher than that of children residing in urban areas. Home deliveries consistently accounted for 53% of deliveries across the antigen-specific datasets, compared with deliveries occurring at health facilities. The distribution of children by sex was balanced, with approximately 50% being female and 50% male across all antigen-specific datasets.

Table 4.1: Study population characteristics across antigen datasets

Characteristic	Antigen					
	BCG N = 14,896	Pentavalent 1 N = 14,291	Pentavalent 2 N = 13,885	Pentavalent 3 N = 13,479	Vitamin A N = 12,195	MCV1 N = 10,837
Residence						
Urban	5,854 (39%)	5,630 (39%)	5,486 (40%)	5,344 (40%)	4,861 (40%)	4,345 (40%)
Rural	9,042 (61%)	8,661 (61%)	8,399 (60%)	8,135 (60%)	7,334 (60%)	6,492 (60%)
Birth order	3.65 (2.38)	3.65 (2.38)	3.65 (2.39)	3.64 (2.38)	3.65 (2.39)	3.64 (2.39)
Place of delivery						
Home	7,969 (53%)	7,637 (53%)	7,424 (53%)	7,186 (53%)	6,499 (53%)	5,729 (53%)
Health facility	6,927 (47%)	6,654 (47%)	6,461 (47%)	6,293 (47%)	5,696 (47%)	5,108 (47%)
Child's gender						
Male	7,512 (50%)	7,205 (50%)	7,029 (51%)	6,824 (51%)	6,179 (51%)	5,512 (51%)
Female	7,384 (50%)	7,086 (50%)	6,856 (49%)	6,655 (49%)	6,016 (49%)	5,325 (49%)
Travel time to nearest health facility						
<30 minutes	11,705 (79%)	11,230 (79%)	10,898 (78%)	10,590 (79%)	9,583 (79%)	8,533 (79%)
31–60 minutes	2,458 (17%)	2,356 (16%)	2,296 (17%)	2,220 (16%)	2,011 (16%)	1,770 (16%)
1–2 hours	573 (3.8%)	551 (3.9%)	540 (3.9%)	523 (3.9%)	469 (3.8%)	421 (3.9%)
>2 hours	160 (1.1%)	154 (1.1%)	151 (1.1%)	146 (1.1%)	132 (1.1%)	113 (1.0%)
Wealth index						
Poorest	3,794 (25%)	3,632 (25%)	3,514 (25%)	3,396 (25%)	3,075 (25%)	2,705 (25%)
Poorer	2,915 (20%)	2,802 (20%)	2,725 (20%)	2,633 (20%)	2,349 (19%)	2,114 (20%)
Middle	2,883 (19%)	2,751 (19%)	2,685 (19%)	2,607 (19%)	2,351 (19%)	2,076 (19%)
Richer	2,836 (19%)	2,728 (19%)	2,653 (19%)	2,586 (19%)	2,350 (19%)	2,085 (19%)
Richest	2,468 (17%)	2,378 (17%)	2,308 (17%)	2,257 (17%)	2,070 (17%)	1,857 (17%)
Maternal education						
No formal education	6,300 (42%)	6,054 (42%)	5,867 (42%)	5,678 (42%)	5,126 (42%)	4,545 (42%)
Primary school	1,638 (11%)	1,574 (11%)	1,533 (11%)	1,496 (11%)	1,348 (11%)	1,191 (11%)
Secondary school	5,241 (35%)	5,007 (35%)	4,877 (35%)	4,730 (35%)	4,297 (35%)	3,830 (35%)
Higher education	1,717 (12%)	1,656 (12%)	1,608 (12%)	1,575 (12%)	1,424 (12%)	1,271 (12%)
Mother's age at birth						
≤19 years	1,259 (8.5%)	1,211 (8.5%)	1,176 (8.5%)	1,148 (8.5%)	1,040 (8.5%)	953 (8.8%)
20–24 years	3,597 (24%)	3,455 (24%)	3,367 (24%)	3,276 (24%)	2,959 (24%)	2,618 (24%)
25–29 years	4,007 (27%)	3,827 (27%)	3,703 (27%)	3,597 (27%)	3,237 (27%)	2,892 (27%)
30–34 years	3,215 (22%)	3,094 (22%)	3,002 (22%)	2,919 (22%)	2,650 (22%)	2,345 (22%)
35–39 years	1,904 (13%)	1,825 (13%)	1,779 (13%)	1,714 (13%)	1,547 (13%)	1,362 (13%)
40–44 years	761 (5.1%)	730 (5.1%)	713 (5.1%)	685 (5.1%)	638 (5.2%)	553 (5.1%)
45 years or older	153 (1.0%)	149 (1.0%)	145 (1.0%)	140 (1.0%)	124 (1.0%)	114 (1.1%)
Geopolitical zone						
North West	4,442 (30%)	4,243 (30%)	4,109 (30%)	3,988 (30%)	3,597 (29%)	3,177 (29%)
North East	3,147 (21%)	3,029 (21%)	2,947 (21%)	2,836 (21%)	2,540 (21%)	2,266 (21%)
North Central	2,726 (18%)	2,625 (18%)	2,545 (18%)	2,465 (18%)	2,219 (18%)	1,968 (18%)
South East	1,633 (11%)	1,572 (11%)	1,529 (11%)	1,493 (11%)	1,349 (11%)	1,218 (11%)
South South	1,520 (10%)	1,459 (10%)	1,421 (10%)	1,386 (10%)	1,274 (10%)	1,143 (11%)
South West	1,428 (9.6%)	1,363 (9.5%)	1,334 (9.6%)	1,311 (9.7%)	1,216 (10.0%)	1,065 (9.8%)

¹ n (%); Mean (SD)

4.2 Heatwave exposure

4.3 Heat Index exposure

4.4 Vaccine timeliness

4.5 Heat exposure and vaccine timeliness

4.6 Timeliness statistics by threshold

5 Discussion

References

1. Ebi KL, Vanos J, Baldwin JW, Bell JE, Hondula DM, Errett NA, et al. Extreme weather and climate change: Population health and health system implications. Annual review of public health [Internet]. 2021 Apr 1;42:293–315. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9013542/>
2. Zhang W, Ding N, Han Y, He J, Zhang N. [The impact of temperature on labor productivity—evidence from temperature-sensitive enterprises](#). Frontiers in Environmental Science. 2023;10.
3. Costello A, Abbas M, Allen A, Ball S, Bell S, Bellamy R, et al. Managing the health effects of climate change: Lancet and university college london institute for global health commission. The Lancet [Internet]. 2009 May 16;373(9676):1693–733. Available from: <https://www.sciencedirect.com/science/article/pii/S0140673609609351>
4. Adrian Simmons HH. Low frequency variability and trends in surface air temperature and humidity from ERA5 and other datasets [Internet]. 2021. Available from: <https://www.ecmwf.int/en/elibrary/81213-low-frequency-variability-and-trends-surface-air-temperature-and-humidity-era5>
5. Change IC. Impacts, adaptation and vulnerability. Contribution of working group II to the sixth assessment report of the intergovernmental panel on climate change. O, Roberts, DC, Tignor, M, Poloczanska, ES, Mintenbeck, K, Alegría, A, Craig, M, Langsdorf, S, Löschke, S, Möller, V, et al, Eds. 2022;3056.
6. Hajat S, Kovats RS, Lachowycz K. Heat-related and cold-related deaths in england and wales: Who is at risk? Occupational and Environmental Medicine [Internet]. 2007 Feb 1;64(2):93–100. Available from: <https://oem.bmj.com/content/64/2/93>

7. Ferron C, Trewick D, Le Conte P, Batard É, Girard L, Potel G. Canicule de l'été 2003: Étude descriptive des décès par coup de chaleur au CHU de Nantes. *La Presse Médicale* [Internet]. 2006 Feb 1;35(2, Part 1):196–9. Available from: <https://www.sciencedirect.com/science/article/pii/S0755498206745535>
8. Stafoggia M, Forastiere F, Agostini D, Caranci N, de'Donato F, Demaria M, et al. Factors affecting in-hospital heat-related mortality: A multi-city case-crossover analysis. *Journal of Epidemiology & Community Health* [Internet]. 2008 Mar 1;62(3):209–15. Available from: <https://jech.bmj.com/content/62/3/209>
9. Hundessa S, Huang W, Xu R, Yang Z, Zhao Q, Gasparrini A, et al. Global excess deaths associated with heatwaves in 2023 and the contribution of human-induced climate change. *The Innovation* [Internet]. 2025 Sept 4;6(10):101110. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC7618246/>
10. Clarke B, Konstantinoudis G, Barnes C, Keeping T, Otto F, Gasparrini A, et al. Climate change tripled heat-related deaths in early summer European heatwave. 2025 July 14; Available from: <https://hdl.handle.net/10044/1/128628>
11. He C, Zhu Y, Guo Y, Bachwenkizi J, Chen R, Kan H, et al. Escalated heatwave mortality risk in sub-Saharan Africa under recent warming trend. *Science Advances* [Internet]. 11(48):eady7379. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12652248/>
12. Curtis S, Fair A, Wistow J, Val DV, Owen K. Impact of extreme weather events and climate change for health and social care systems. *Environmental Health* [Internet]. 2017 Dec 5;16(1):128. Available from: <https://doi.org/10.1186/s12940-017-0324-3>
13. Carmichael C, Bickler G, Kovats S, Pencheon D, Murray V, West C, et al. Overheating and hospitals - what do we know? *Journal of Hospital Administration* [Internet]. 2012 Oct 11;2(1):1. Available from: <https://www.sciencedirect.com/journal/index.php/jha/article/view/1651>

14. ADEFISAN EA, AHMAD S. ASSESSMENT OF HEAT WAVE EVENTS IN a CHANGING CLIMATE OVER NIGERIA. *International journal of research science and management*. 2018;5(8):115133.
15. Morakinyo TE, Ishola KA, Eresanya EO, Daramola MT, Balogun IA. Spatio-temporal characteristics of heat stress over nigeria using evaluated ERA5-HEAT reanalysis data. *Weather and Climate Extremes* [Internet]. 2024 Sept 1;45:100704. Available from: <https://www.sciencedirect.com/science/article/pii/S2212094724000653>
16. Climate change-induced heatwaves in nigeria: Causes, challenges, and adaptive strategies. *Journal of Environmental Management* [Internet]. 2025 Nov 1;394:127433. Available from: <https://www.sciencedirect.com/science/article/pii/S0301479725034097>
17. Oluwaseun OA. Heatwave in nigeria: Then and now – a critical analysis [Internet]. 2024. Available from: <https://dailytrust.com/heatwave-in-nigeria-then-and-now-a-critical-analysis/>
18. Gbadegesin AS, Olorunfemi FB, Raheem UA. Urban vulnerability to climate change and natural hazards in nigeria. In: Brauch HG, Oswald Spring Ú, Mesjasz C, Grin J, Kameri-Mbote P, Chourou B, et al., editors. *Berlin, Heidelberg: Springer*; 2011. p. 669–87. Available from: https://doi.org/10.1007/978-3-642-17776-7_39
19. Federal Ministry of Health and Social Welfare of Nigeria (FMoHSW), National Population Commission (NPC) [Nigeria], ICF. *Nigeria demographic and health survey 2024* [Internet]. Abuja, Nigeria; Rockville, Maryland, USA; 2025. Available from: <https://dhsprogram.com/pubs/pdf/FR395/FR395.pdf>
20. Williams SV, Akande T, Abbas K. Systematic review of social determinants of childhood immunisation in low- and middle-income countries and equity impact analysis of childhood vaccination coverage in nigeria. *PLOS ONE* [Internet]. 2024 Mar 6;19(3):e0297326. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10917251/>
21. Ogunniyi TJ. [Revealing the challenges and efforts of routine immunization coverage in Nigeria](#). *Health Science Reports*. 2024 Aug;7(8):e70000.

22. Ophori EA, Tula MY, Azih AV, Okojie R, Ikpo PE. Current trends of immunization in nigeria: Prospect and challenges. *Tropical Medicine and Health* [Internet]. 2014 June;42(2):67–75. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC4139536/>

23. Alabi MA, Fasasi MI, Obiora RU, Nwankwo GI, Ukwu HU. Factors associated with full childhood vaccination coverage among young mothers in northern nigeria. *PAMJ Clinical Medicine* [Internet]. 2024 Jan 4;47(4). Available from: <https://www.panafrican-med-journal.com//content/article/47/4/full>

24. Salako J, Bakare D, Uchendu OC, Bakare AA, Graham H, Falade AG. Factors associated with immunization status among children aged 12-59 months in lagelu local government area, ibadan: A cross-sectional study. *PAMJ Clinical Medicine* [Internet]. 2024 Jan 30;47(35). Available from: <https://www.panafrican-med-journal.com//content/article/47/35/full>

25. Watch NH. Closing the gap: Partnering for 'the big catch-up' on routine immunisation in nigeria [Internet]. 2023. Available from: <https://nigeriahealthwatch.com/articles/thought-leadership/closing-the-gap-partnering-for-the-big-catch-up-on-routine-immunisation-in-nigeria/>

26. Adegoke AA, Balogun FM. How the COVID-19 pandemic affected infant vaccination trends in rural and urban communities in ibadan, nigeria: A cross-sectional study. 2024 July 1; Available from: https://bmjopen.bmj.com/content/14/7/e073272?utm_medium=reward&utm_source=trendmd

27. Umeokonkwo AA, Ezeonu TC, Agu AP, Umeokonkwo CD. Timeliness of vaccination among children attending immunization clinic at a tertiary hospital in south-east nigeria. *Nigerian Medical Journal : Journal of the Nigeria Medical Association* [Internet]. 62(4):153–61. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11058447/>

28. Derqui N, Blake IM, Gray EJ, Cooper LV, Grassly NC, Pons-Salort M, et al. Timeliness of 24 childhood immunisations and evolution of vaccination delay: Analysis of data from 54 low- and middle-income countries. *PLOS Global Public Health* [Internet]. 2024 Nov 26;4(11):e0003749. Available from: <https://journals.plos.org/globalpublichealth/article?id=10.1371/journal.pgph.0003749>

29. MacDonald SE, Russell ML, Liu XC, Simmonds KA, Lorenzetti DL, Sharpe H, et al. Are we speaking the same language? An argument for the consistent use of terminology and definitions for childhood vaccination indicators. *Human Vaccines & Immunotherapeutics* [Internet]. 2018 Dec 20;15(3):740–7. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6605715/>

30. Hughes SL, Bolotin S, Khan S, Li Y, Johnson C, Friedman L, et al. The effect of time since measles vaccination and age at first dose on measles vaccine effectiveness – a systematic review. *Vaccine* [Internet]. 2020 Jan 16;38(3):460–9. Available from: <https://www.sciencedirect.com/science/article/pii/S0264410X19314793>

31. Goult E, Barrero Guevara LA, Briga M, Domenech de Cellès M. Estimating the optimal age for infant measles vaccination. *Nature Communications* [Internet]. 2024 Nov 15;15:9919. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11568136/>

32. Guerra FA. Delays in immunization have potentially serious health consequences. *Pediatric Drugs* [Internet]. 2007 May 1;9(3):143–8. Available from: <https://doi.org/10.2165/00148581-200709030-00002>

33. Wariri O, Okomo U, Kwarshak YK, Utazi CE, Murray K, Grundy C, et al. Timeliness of routine childhood vaccination in 103 low-and middle-income countries, 1978–2021: A scoping review to map measurement and methodological gaps. *PLOS Global Public Health* [Internet]. 2022 July 14;2(7):e0000325. Available from: <https://journals.plos.org/globalpublichealth/article?id=10.1371/journal.pgph.0000325>

34. Clark AD, Feikin DR, Danovaro-Holliday MC, Sanderson CFB. Timeliness of children's vaccinations in 91 low-income and middle-income countries: An analysis of survey data. *The Lancet Global Health* [Internet]. 2026 May 1;14(5):e714–22. Available from: <https://www.sciencedirect.com/science/article/pii/S2214109X25005546>
35. Masters NB, Wagner AL, Boulton ML. Vaccination timeliness and delay in low- and middle-income countries: A systematic review of the literature, 2007-2017. *Human Vaccines & Immunotherapeutics* [Internet]. 2019 Dec 2;15(12):2790–805. Available from: <https://doi.org/10.1080/21645515.2019.1616503>
36. Siddiqi DA, Iftikhar S, Anfossi CM, Siddique M, Shah MT, Dharma VK, et al. Assessing the impact of heat waves on childhood immunization coverage in sindh, pakistan: Insights from 132.4 million doses recorded in the provincial electronic immunization registry (2018–2024). *Vaccine* [Internet]. 2025 Aug 13;61:127424. Available from: <https://www.sciencedirect.com/science/article/pii/S0264410X25007212>
37. Zheng H, Zhu Y, Chu H, Xu Z, Jit M, Ding Z, et al. [Association between ambient temperature and childhood vaccination coverage in low- and middle-income countries](#). *BMC medicine*. 2026 Mar 11;24(1):296.
38. Akmatov MK, Mikolajczyk RT. Timeliness of childhood vaccinations in 31 low and middle-income countries. *J Epidemiol Community Health* [Internet]. 2012 July 1;66(7):e14–4. Available from: <https://jech.bmj.com/content/66/7/e14>
39. Bassoum O, Kimura M, Dia AT, Lemoine M, Shimakawa Y. Coverage and timeliness of birth dose vaccination in sub-saharan africa: A systematic review and meta-analysis. *Vaccines* [Internet]. 2020 June 10;8(2). Available from: <https://www.mdpi.com/2076-393X/8/2/301>
40. Clark A, Sanderson C. Timing of children's vaccinations in 45 low-income and middle-income countries: An analysis of survey data. *The Lancet* [Internet]. 2009 May 2;373(9674):1543–9. Available from: <https://www.sciencedirect.com/science/article/pii/S0140673609603172>

41. Janusz CB, Frye M, Mutua MK, Wagner AL, Banerjee M, Boulton ML. Vaccine delay and its association with undervaccination in children in sub-saharan africa. *American Journal of Preventive Medicine* [Internet]. 2021 Jan 1;60(1, Supplement 1):S53–64. Available from: <https://www.sciencedirect.com/science/article/pii/S0749379720304268>
42. Mutua MK, Mohamed SF, Porth JM, Faye CM. Inequities in on-time childhood vaccination: Evidence from sub-saharan africa. *American Journal of Preventive Medicine* [Internet]. 2021 Jan 1;60(1, Supplement 1):S11–23. Available from: <https://www.sciencedirect.com/science/article/pii/S0749379720304256>
43. Jain R, Chopra A, Falézan C, Patel M, Dupas P. COVID-19 related immunization disruptions in rajasthan, india: A retrospective observational study. *Vaccine* [Internet]. 2021 July 13;39(31):4343–50. Available from: <https://www.sciencedirect.com/science/article/pii/S0264410X21007581>
44. Van Oldenborgh GJ, Wehner MF, Vautard R, Otto FEL, Seneviratne SI, Stott PA, et al. Attributing and projecting heatwaves is hard: We can do better. *Earth's Future* [Internet]. 2022;10(6):e2021EF002271. Available from: <https://onlinelibrary.wiley.com/doi/abs/10.1029/2021EF002271>
45. Faye M, Dème A, Diongue AK, Diouf I. Impact of different heat wave definitions on daily mortality in bandafassi, senegal. *PLOS ONE* [Internet]. 2021 Apr 5;16(4):e0249199. Available from: <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0249199>
46. Philip S, Kew S, Oldenborgh GJ van, Otto F, Vautard R, Wiel K van der, et al. A protocol for probabilistic extreme event attribution analyses. *Advances in Statistical Climatology, Meteorology and Oceanography* [Internet]. 2020 Nov 10;6(2):177–203. Available from: <https://ascmo.copernicus.org/articles/6/177/2020/>
47. Perkins SE. A review on the scientific understanding of heatwaves—their measurement, driving mechanisms, and changes at the global scale. *Atmospheric Research* [Internet]. 2015 Oct 1;164-165:242–67. Available from: <https://www.sciencedirect.com/science/article/pii/S0169809515001738>

48. Perkins SE, Alexander LV. On the measurement of heat waves. *Journal of Climate* [Internet]. 2013 July 1;26(13):4500–17. Available from: <https://journals.ametsoc.org/view/journals/clim/26/13/jcli-d-12-00383.1.xml>
49. Stott PA, Christidis N, Otto FEL, Sun Y, Vanderlinden JP, Oldenborgh GJ van, et al. Attribution of extreme weather and climate-related events. *WIREs Climate Change* [Internet]. 2016;7(1):23–41. Available from: <https://onlinelibrary.wiley.com/doi/abs/10.1002/wcc.380>
50. Fonjong L, Matose F, Sonnenfeld DA. Climate change in africa: Impacts, adaptation, and policy responses. *Global Environmental Change* [Internet]. 2024 Dec 1;89:102912. Available from: <https://www.sciencedirect.com/science/article/pii/S095937802400116X>
51. Schmidt S, Zhang Y, Li W, Li H. Urban morphology and the heat island effect in african cities: Evidence from ghana, togo, and tanzania. *Remote Sensing Applications: Society and Environment* [Internet]. 2026 Apr 1;42:102005. Available from: <https://www.sciencedirect.com/science/article/pii/S2352938526001382>
52. Sapari H, Selamat MI, Isa MR, Ismail R, Mahiyuddin WRW. The impact of heat waves on health care services in low- or middle-income countries: Protocol for a systematic review. *JMIR Research Protocols* [Internet]. 2023 Oct 16;12(1):e44702. Available from: <https://www.researchprotocols.org/2023/1/e44702>
53. De Boeck K, Decouttere C, Vandaele N. Vaccine distribution chains in low- and middle-income countries: A literature review. *Omega* [Internet]. 2020 Dec 1;97:102097. Available from: <https://www.sciencedirect.com/science/article/pii/S0305048319304098>
54. Jadeja N, Omumbo J, Adelekan I, Rees H, Bonfoh B, Kariuki T, et al. Climate and health strategies must take vaccination into account. *Nature Microbiology* [Internet]. 2023 Dec;8(12):2215–6. Available from: <https://www.nature.com/articles/s41564-023-01537-1>

55. Nagata JM, Epstein A, Ganson KT, Benmarhnia T, Weiser SD. Drought and child vaccination coverage in 22 countries in sub-saharan africa: A retrospective analysis of national survey data from 2011 to 2019. PLoS Medicine [Internet]. 2021 Sept 28;18(9):e1003678. Available from: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8478213/>
56. Hersbach H, Bell B, Berrisford P, Biavati G, Horányi A, Muñoz Sabater J, et al. ERA5 hourly data on single levels from 1940 to present. 2023; Available from: <https://doi.org/10.24381/cds.adbb2d47>
57. Copernicus Climate Change Service. ERA5 hourly data on single levels from 1940 to present. 2023; Available from: <https://doi.org/10.24381/cds.adbb2d47>
58. Hufkens K, Stauffer R, Campitelli E. The ecwmfr package: An interface to ECMWF API endpoints. 2019; Available from: <https://bluegreen-labs.github.io/ecmwfr/>
59. World Health Organization, Biostat Global Consulting. VCQI user's guide [Internet]. 2021. Available from: <https://www.biostatglobal.com/downloads/VCQI%20User's%20Guide.pdf>
60. R. A. Rigby, D. M. Stasinopoulos. Generalized additive models for location, scale and shape,(with discussion). Applied Statistics. 2005;54:507–54.
61. Northrop PJ, Ji J. Gamlssx: Generalized additive extreme value models for location, scale and shape [Internet]. 2025. Available from: <https://CRAN.R-project.org/package=gamlssx>
62. Rigby RA, Stasinopoulos MD, Heller GZ, Bastiani FD. [Distributions for modeling location, scale, and shape: Using GAMLSS in r](#). New York: Chapman; Hall/CRC; 2019.
63. Anderson GB, Bell ML, Peng RD. [Methods to calculate the heat index as an exposure metric in environmental health research](#). Environmental Health Perspectives. 2013 Oct;121(10):1111–9.

64. National Weather Service. Meteorological conversions and calculations: Heat index calculator [Internet]. 2011. Available from: https://www.wpc.ncep.noaa.gov/html/heatindex_equation.shtml
65. R Core Team. R: A language and environment for statistical computing [Internet]. Vienna, Austria: R Foundation for Statistical Computing; 2025. Available from: <https://www.R-project.org/>
66. Wickham H, Averick M, Bryan J, Chang W, McGowan LD, François R, et al. Welcome to the tidyverse. Journal of Open Source Software [Internet]. 2019 Nov 21;4(43):1686. Available from: <https://joss.theoj.org/papers/10.21105/joss.01686>
67. Hufkens K, Stauffer R, Campitelli E. The ecwmfr package: An interface to ECMWF API endpoints. 2019; Available from: <https://bluegreen-labs.github.io/ecmwfr/>
68. Wickham H, Miller E, Smith D. Haven: Import and export 'SPSS', 'stata' and 'SAS' files [Internet]. 2025. Available from: <https://CRAN.R-project.org/package=haven>
69. Hijmans RJ. Terra: Spatial data analysis [Internet]. 2026. Available from: <https://CRAN.R-project.org/package=terra>
70. Pebesma E, Bivand R. [Spatial data science: With applications in r](#). New York: Chapman; Hall/CRC; 2023.
71. Northrop PJ, Ji J. Gamlssx: Generalized additive extreme value models for location, scale and shape [Internet]. 2025. Available from: <https://CRAN.R-project.org/package=gamlssx>
72. R. A. Rigby, D. M. Stasinopoulos. Generalized additive models for location, scale and shape,(with discussion). Applied Statistics. 2005;54:507–54.
73. Lumley T. Analysis of complex survey samples. Journal of Statistical Software. 2004;9(1):1–19.

74. Lumley T. Complex surveys: A guide to analysis using r: A guide to analysis using r. John Wiley; Sons; 2010.
75. Allaire JJ, Xie Y, Dervieux C, McPherson J, Luraschi J, Ushey K, et al. Rmarkdown: Dynamic documents for r [Internet]. 2026. Available from: <https://github.com/rstudio/rmarkdown>
76. Zhu H. kableExtra: Construct complex table with 'kable' and pipe syntax [Internet]. 2024. Available from: <https://CRAN.R-project.org/package=kableExtra>

Appendix